

Post Gold

Today's modern day athlete demands more from cutting-edge science and expects a post-workout to deliver a faster, stronger, leaner recovery. Although protein is an essential part of post workout recovery, it is just one piece of the puzzle that our bodies need. **Post Gold** has the rest and it brings a whole new meaning to the word GAINS! Key include **BCAAs, Beta Alanine, Betaine, GABA, Glutamine, and Choline Bitartrate**.

The BCAAs: leucine, isoleucine, and valine, make up 35-40% of the essential dietary amino acids in body protein and 14-18% of total amino acids in muscle protein.^{1,2} Based on these percentages, it is no surprise that **BCAAs have become an essential part of any athlete's daily routine**. Recent studies have shown the promising effects of ingesting BCAAs to reduce muscle damage, fatigue, and soreness, as well as increase energy metabolism, and enhance overall exercise performance.³⁻⁵ In addition to BCAAs, Beta Alanine is another amino acid that has gained tremendous popularity as a vital component to any muscle building and recovery matrix. **Beta Alanine** has been researched extensively and literature shows **significant decreases in perceived exertion, increases in muscle carnosine levels, and improvements in muscular endurance**.^{6,7}

Betaine is an organic compound that is found naturally in a variety of food sources including sugar beets, wheat bran, spinach, shrimp, and much more.⁸ Its main function is to **protect cells under stress** and has also been shown to **increase growth hormone levels by approximately 6.37% and IGF-1 levels by 11.6%**, thus enhancing the anabolic signaling environment in the body and protein synthesis in muscle.^{8,9} In addition, **GABA**, another revolutionary ingredient found in **Post Gold**, has gained increasing popularity for its effect on growth hormone levels. GABA (Gamma Amino Butyric Acid) is a neurotransmitter with the primary role of sending chemical messages throughout the brain and nervous system. Studies have shown that GABA supplementation at rest and post exercise **increases growth hormone levels by 400%**, therefore playing an extremely important role in the growth and maintenance of skeletal muscle.^{10,11}

Glutamine, another key ingredient in **Post Gold**, has shown great promise for increasing growth hormone levels. Studies have found that supplementation with **Glutamine resulted in an astounding 430% increase in plasma growth hormone**.¹² The final piece to this recovery puzzle is **Choline, a member of the vitamin B-Complex family**. Supplementing with Choline can **minimize** the development of exercise-related **fatigue** since plasma choline levels are significantly reduced following intense exercise. Choline supplementation also contributes to a **significant increase in lipid metabolism** and studies have shown as much as a **10.23% reduction in body fat**.¹³

These ingredients are backed by innovative and ground-breaking research, delivering a formula designed to take recovery and muscle gains to the NEXT LEVEL! Recover FASTER, STRONGER, and LEANER with this amazing Post Workout Recovery Matrix from Ultimate Nutrition's Gold Series! **Go after the Gold with Post Gold!!!**

1. Riazi R, Wykes LJ, Ball RO, Pencharz PB. The total branched-chain amino acid requirement in young healthy adult men determined by indicator amino acid oxidation by use of L-[1-13C]phenylalanine. J Nutr. 2003;133:1383-9.

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2. Layman DK, Baum JI. Dietary protein impact on glycemic control during weight loss. *J Nutr.* 2004;134 (Suppl):S968–73.
3. Kim, D. H., Kim, S. H., Jeong, W. S., & Lee, H. Y. (2013). Effect of BCAA intake during endurance exercises on fatigue substances, muscle damage substances, and energy metabolism substances. *Journal of Exercise Nutrition and Biochemistry*, 17(4), 169-180.
4. Leahy, D. T., & Pintauro, S. J. (2013). Branched-chain amino acid plus glucose supplement reduces exercise-induced delayed onset muscle soreness in college-age females. *ISRN nutrition*, 2013.
5. Shimomura, Y., Yamamoto, Y., Bajotto, G., Sato, J., Murakami, T., Shimomura, N., ... & Mawatari, K. (2006). Nutraceutical effects of branched-chain amino acids on skeletal muscle. *The Journal of nutrition*, 136(2), 529S-532S.
6. Glenn, J. M., Smith, K., Moyon, N. E., Binns, A., & Gray, M. (2015). Effects of acute beta-alanine supplementation on anaerobic performance in trained female cyclists. *Journal of nutritional science and vitaminology*, 61(2), 161-166.
7. Trexler, E. T., Smith-Ryan, A. E., Stout, J. R., Hoffman, J. R., Wilborn, C. D., Sale, C., ... & Campbell, B. (2015). International society of sports nutrition position stand: Beta-Alanine. *Journal of the International Society of Sports Nutrition*, 12(1), 1.
8. Apicella, J. M., Lee, E. C., Bailey, B. L., Saenz, C., Anderson, J. M., Craig, S. A., ... & Maresh, C. M. (2013). Betaine supplementation enhances anabolic endocrine and Akt signaling in response to acute bouts of exercise. *European journal of applied physiology*, 113(3), 793-802.
9. Pryor, J. L., Craig, S. A., & Swensen, T. (2012). Effect of betaine supplementation on cycling sprint performance. *Journal of the International Society of Sports Nutrition*, 9(1), 1.
10. Powers, M. E., Yarrow, J. F., McCoy, S. C., & Borst, S. E. (2008). Growth hormone isoform responses to GABA ingestion at rest and after exercise. *Medicine and science in sports and exercise*, 40(1), 104.
11. Cavagnini, F., Invitti, C., Pinto, M., Maraschini, C., Di Landro, A., Dubini, A., & Marelli, A. (1980). Effect of acute and repeated administration of gamma aminobutyric acid (GABA) on growth hormone and prolactin secretion in man. *Acta endocrinologica*, 93(2), 149-154.
12. Welbourne, T. C. (1995). Increased plasma bicarbonate and growth hormone after an oral glutamine load. *The American journal of clinical nutrition*, 61(5), 1058-1061.
13. Elsayy, G., Abdelrahman, O., & Hamza, A. (2014). Effect of choline supplementation on rapid weight loss and biochemical variables among female Taekwondo and Judo athletes. *Journal of human kinetics*, 40(1), 77-82.